# Responsibility: the law, the disclosure, and your name on it

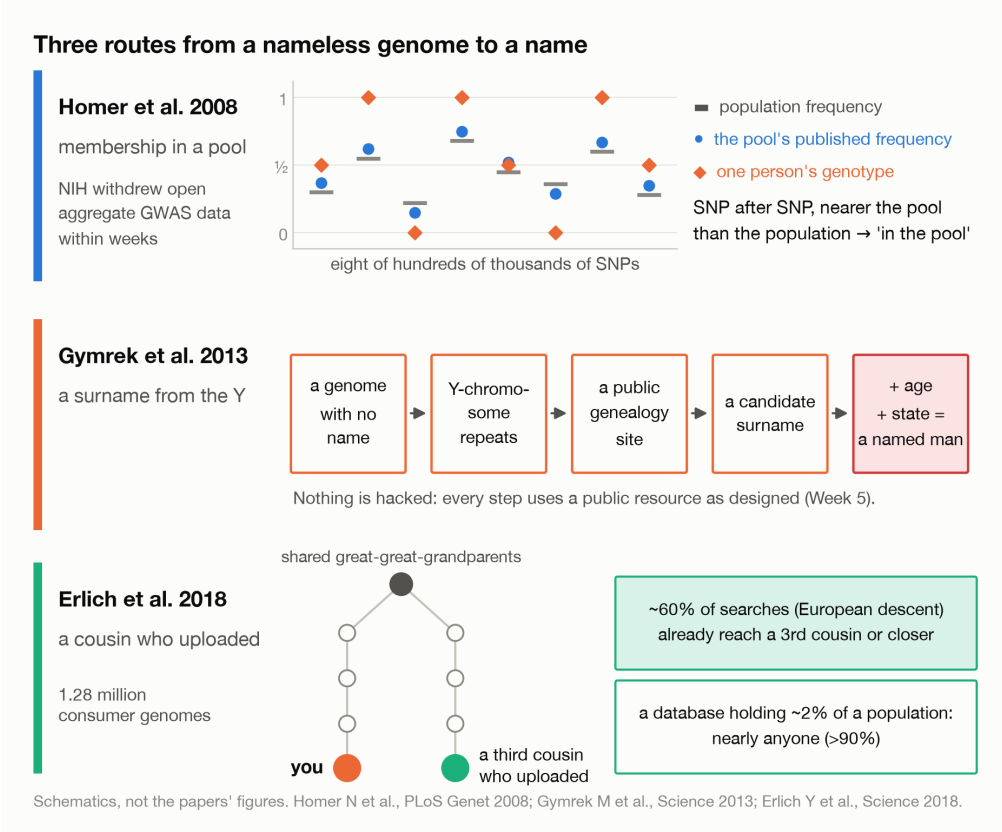
- Sorumluluk: hukuk, beyan ve altındaki imzanız
- BIB Week 14

# Ad of the Week #14

- Skills Ledger: tick data protection and documented, disclosed analysis

# Genomic data cannot be anonymised, and is never only yours

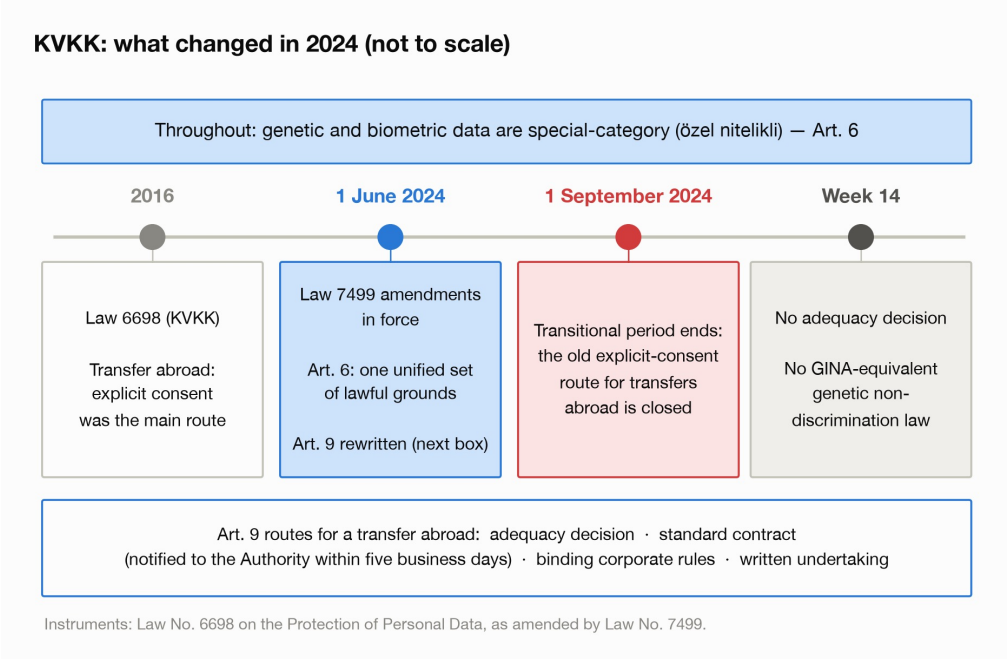
- Homer 2008: a pool's published allele frequencies reveal whether one person's DNA is in it
  - NIH pulled open aggregate (toplu) GWAS data within weeks: aggregates are not safe by default
- Gymrek 2013: Y-chromosome repeats + a genealogy site → surname → identity (Week 5)
- Erlich 2018: ~60% of searches (European descent) already reach a third cousin or closer
  - a database holding ~2% of a population reaches a third cousin for nearly anyone (>90%)
- Half your genome is each parent's, pieces are every cousin's: their consent (rıza) was never asked




*Three routes from a nameless genome to a name — schematics, after Homer 2008, Gymrek 2013 and Erlich 2018*

# KVKK after Law 7499: the law in six sentences

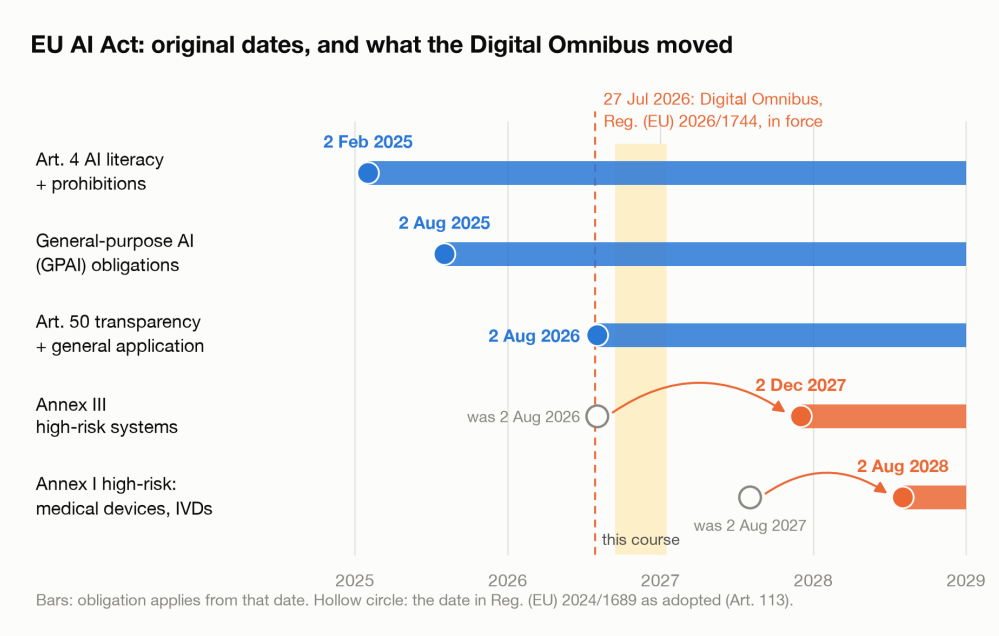
- KVKK, Law 6698, as amended by Law 7499: the amendments are in force since 1 June 2024
- Genetic and biometric data are special-category (özel nitelikli) personal data
- Art. 6: one unified set of lawful grounds now covers every special category
- Art. 9: transfer abroad (yurt dışına aktarım) needs adequacy, a standard contract, BCRs or an undertaking
- The transitional explicit-consent (açık rıza) route for transfers abroad closed on 1 September 2024
- Türkiye has no adequacy decision and no GINA-equivalent genetic non-discrimination law




*KVKK: what changed in 2024 — not to scale*

# The EU AI Act: the corrected timeline

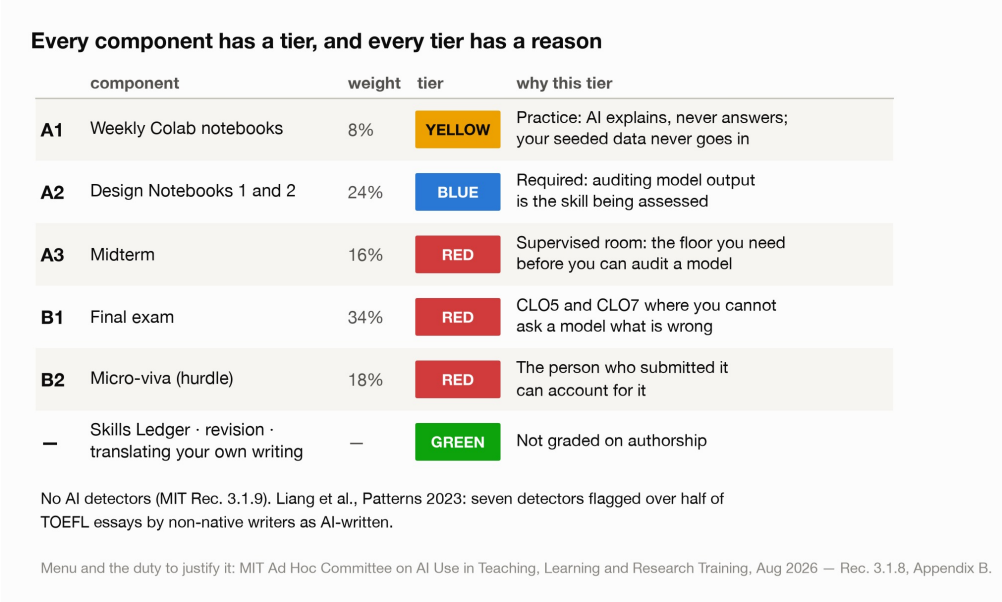
- Reg. (EU) 2024/1689: Art. 4 AI literacy (YZ okuryazarlığı) and prohibitions apply from 2 Feb 2025
- General-purpose AI (GPAI) obligations: 2 August 2025
- Art. 50 transparency obligations and general application: 2 August 2026
- Digital Omnibus, Regulation (EU) 2026/1744, in force 27 July 2026, postponed the high-risk rules:
  - Annex III high-risk (yüksek riskli) → 2 Dec 2027; Annex I: medical devices, IVDs → 2 Aug 2028
- A slide written before August 2026 is likely wrong on exactly this line: check the instrument




*EU AI Act: original dates, and what the Digital Omnibus moved*

# Article 4 in this room: the course AI policy

- A1 weekly notebooks — YELLOW: AI explains, it does not answer; outsourced practice is no practice
- A2 Design Notebooks — BLUE: AI use required and the record submitted; the audit is the skill
- A3 midterm, B1 final, B2 micro-viva — RED: prohibition is only real in a supervised room
- Each tier states its reason — the menu and the duty to justify it: MIT report (2026), Rec. 3.1.8
- No AI detectors (MIT Rec. 3.1.9). Liang 2023: seven flagged over half of TOEFL essays as AI-written
- A concern needs checkable evidence — your seed, a re-run, the viva — never a detector's score



**Every component has a tier, and every tier has a reason**

| | component | weight | tier | why this tier |
|---|---|---|---|---|
| A1 | Weekly Colab notebooks | 8% | YELLOW | Practice: AI explains, never answers; your seeded data never goes in |
| A2 | Design Notebooks 1 and 2 | 24% | BLUE | Required: auditing model output is the skill being assessed |
| A3 | Midterm | 16% | RED | Supervised room: the floor you need before you can audit a model |
| B1 | Final exam | 34% | RED | CLO5 and CLO7 where you cannot ask a model what is wrong |
| B2 | Micro-viva (hurdle) | 18% | RED | The person who submitted it can account for it |
| – | Skills Ledger · revision · translating your own writing | – | GREEN | Not graded on authorship |

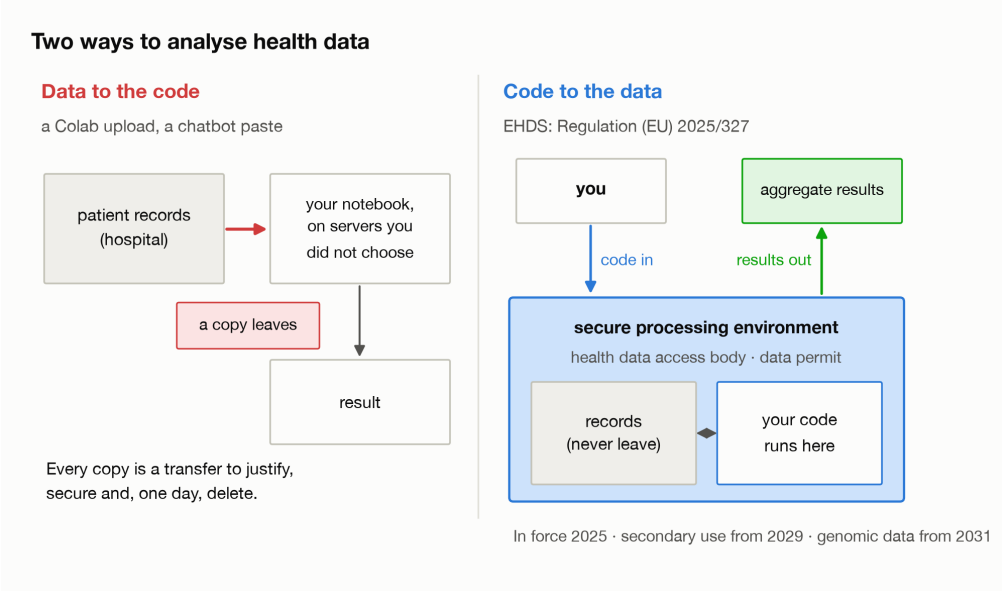
No AI detectors (MIT Rec. 3.1.9). Liang et al., Patterns 2023: seven detectors flagged over half of TOEFL essays by non-native writers as AI-written.

Menu and the duty to justify it: MIT Ad Hoc Committee on AI Use in Teaching, Learning and Research Training, Aug 2026 — Rec. 3.1.8, Appendix B.

*The tier and the reason for every assessed component; GREEN for the Skills Ledger*

# EHDS: bring the code to the data

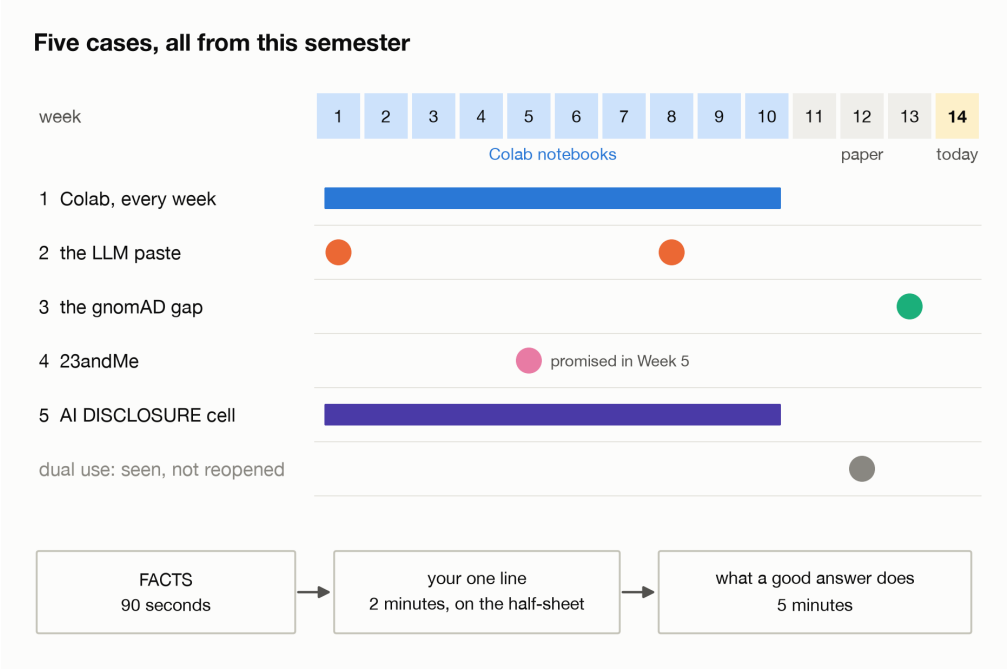
- European Health Data Space, Regulation (EU) 2025/327 (Avrupa Sağlık Veri Alanı) — in force 2025
- Secondary use — research, innovation, policy — applies from 2029; genomic data deferred to 2031
- Access is by data permit from a national health data access body, not by download
- The analysis runs in a secure processing environment: results leave, records do not
- The structural lesson outlives every date: bring the code to the data
- Hold this for Cases 1 and 2: the safest transfer is the one that never happens




*Data to the code versus code to the data — schematic*

# The case clinic: five things this class actually did

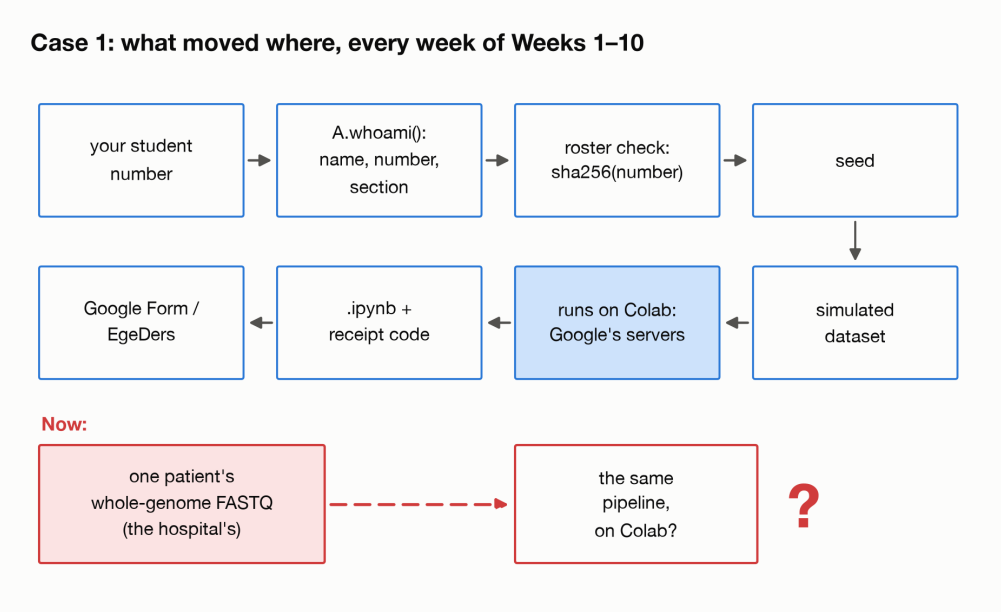
- Five cases, nine minutes each, taken from your own semester — not from a textbook
- Facts (90 s) → your one line, in its box on the half-sheet (2 min) → what a good answer does (5 min)
- Not a moral supplement: every case is a methods question with a defensible answer
- 1 Colab, every week · 2 the LLM paste, Weeks 1 and 8 · 3 the gnomAD gap, Week 13
- 4 23andMe, promised in Week 5 · 5 your AI DISCLOSURE cell, in every notebook
- Dual use (Week 12: redesigned toxins passed DNA-synthesis screening): seen, and not reopened today




*Five cases, all from this semester, and the format each one runs in*

# Case 1 · Your notebooks ran on Colab, every week

- Weeks 1–10: every notebook ran on Colab — Google's servers — with your name and student number in it
- Your dataset was simulated from your student number: no human data, ever (Week 5)
- The course roster stores your student number as a sha256 hash (Week 1)
- Now a hospital asks you to run the same pipeline on one patient's whole-genome FASTQ
- Question: was any of your semester's data personal data — and what changes with the patient's FASTQ?




*Case 1: what moved where, every week of Weeks 1–10 — and the new request*

# Case 1 · What a good answer does

- Sorts the objects: simulated data is nobody's; name + number is personal data (kişisel veri)
- The hash is only pseudonymisation (takma adlandırma): hash every possible number and match
- Classes the FASTQ: genetic, so special-category — it names the patient and their relatives
- Names the transfer: you cannot choose where Colab runs → treat it as transfer abroad (Art. 9)
- Names a lawful route: an Art. 6 ground plus an Art. 9 mechanism — or no transfer: run it on site
- Week 5's twist: an embedding of that genome is still that genome

**Case 1: four objects, three questions, in this order**

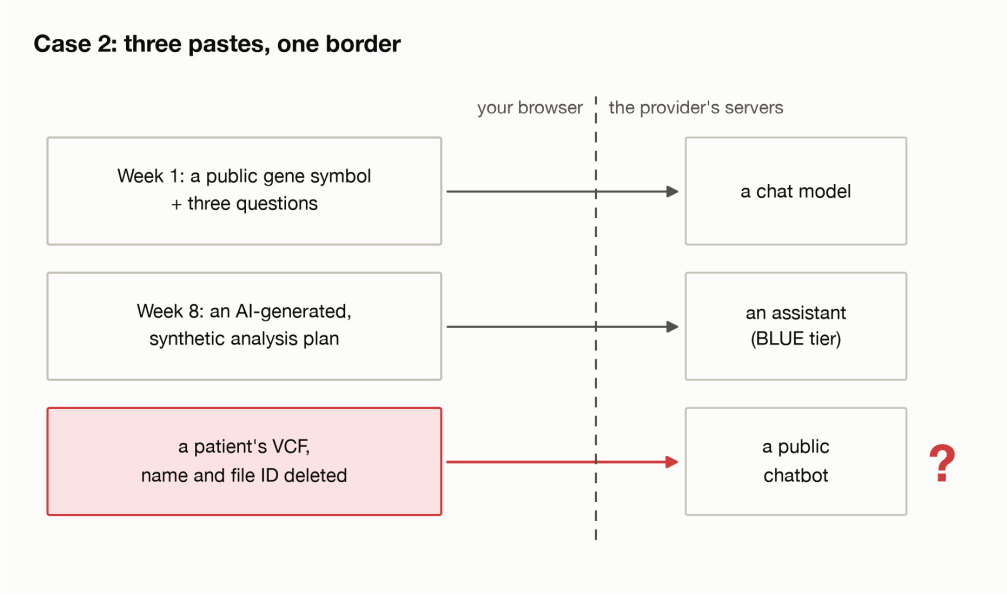
| object | personal data? | what it is |
|---|---|---|
| simulated, seeded dataset | **no** | nobody's genome — a design choice (Week 5) |
| name + student number | **yes** | ordinary personal data (kişisel veri) |
| sha256(student number) | **yes** | pseudonymised: hash every possible number and match — still personal |
| a patient's FASTQ | **yes — special category** | genetic: names the patient and their relatives; Art. 6 ground + Art. 9 route, or no transfer at all |

**① personal at all? ② special category? ③ does it leave the country?**

*Four objects, three questions, asked in this order*

# Case 2 · The paste

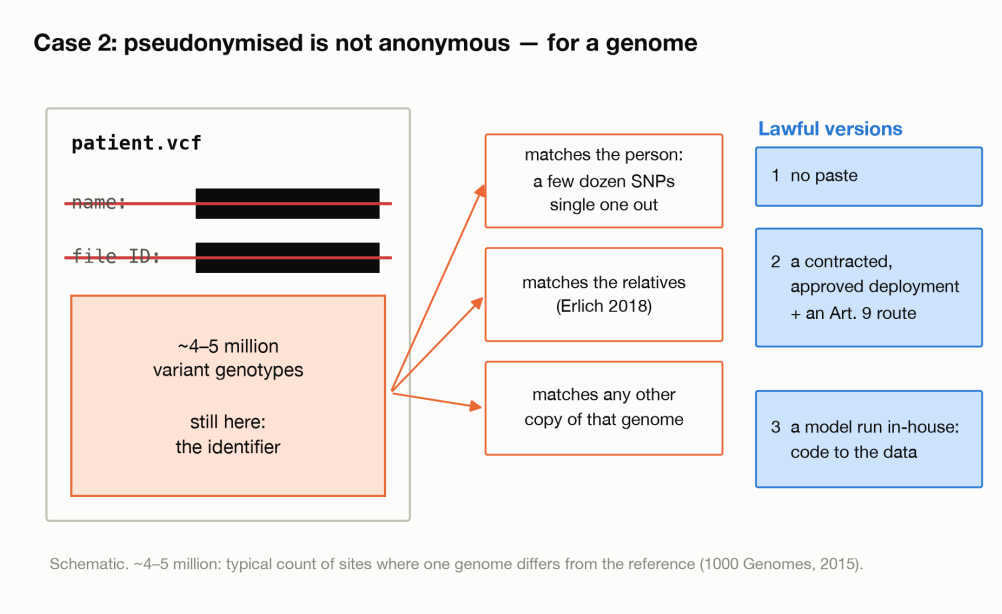
- Week 1: you pasted questions about your gene into a chat model, then checked it against GENCODE
- Week 8: you used an assistant on an AI-generated analysis plan — required, at the BLUE tier
- The YELLOW rule all term: your seeded data never goes into a model
- Now: a colleague deletes the name and file ID from a patient's VCF and pastes it into a public chatbot
- Question: which step was the transfer — and would pseudonymisation have helped?




*Case 2: three pastes, one border — schematic*

# Case 2 · What a good answer does

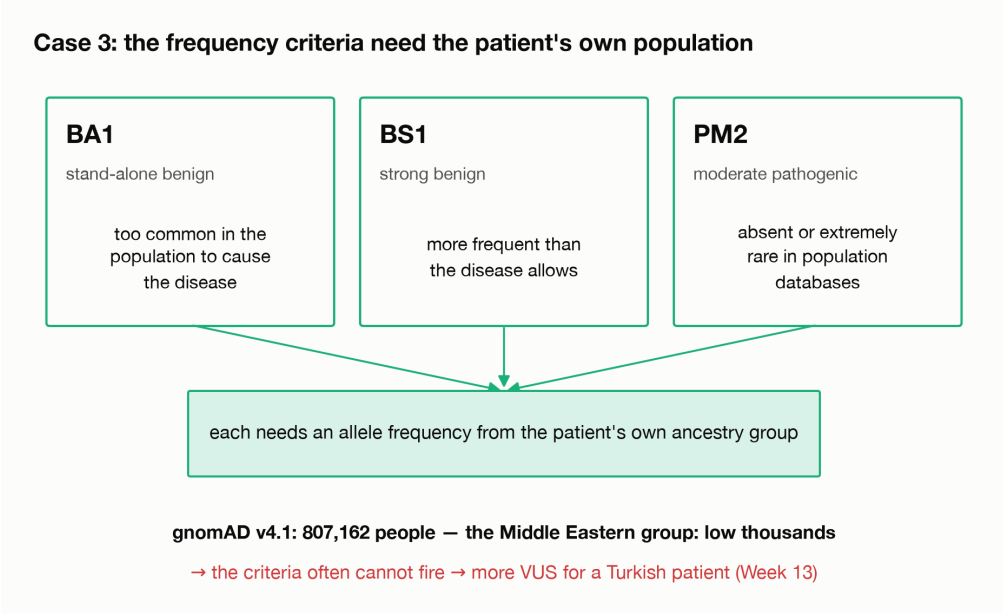
- Names the step: pressing Enter — disclosure to a third party, usually abroad: processing and transfer
- Checks the semester: a public gene symbol and a synthetic plan are not personal data — by design
- Breaks pseudonymisation: delete the name and the genotypes remain — they are the identifier
- Knows a paste cannot be recalled: the provider may retain inputs, and its terms decide, not you
- Lawful versions: no paste; a contracted, approved deployment with an Art. 9 route; a model run in-house
- Only anonymous data falls outside KVKK — and a genome cannot be made anonymous




*Pseudonymised is not anonymous, for a genome — schematic*

# Case 3 · Who should own a Turkish population reference?

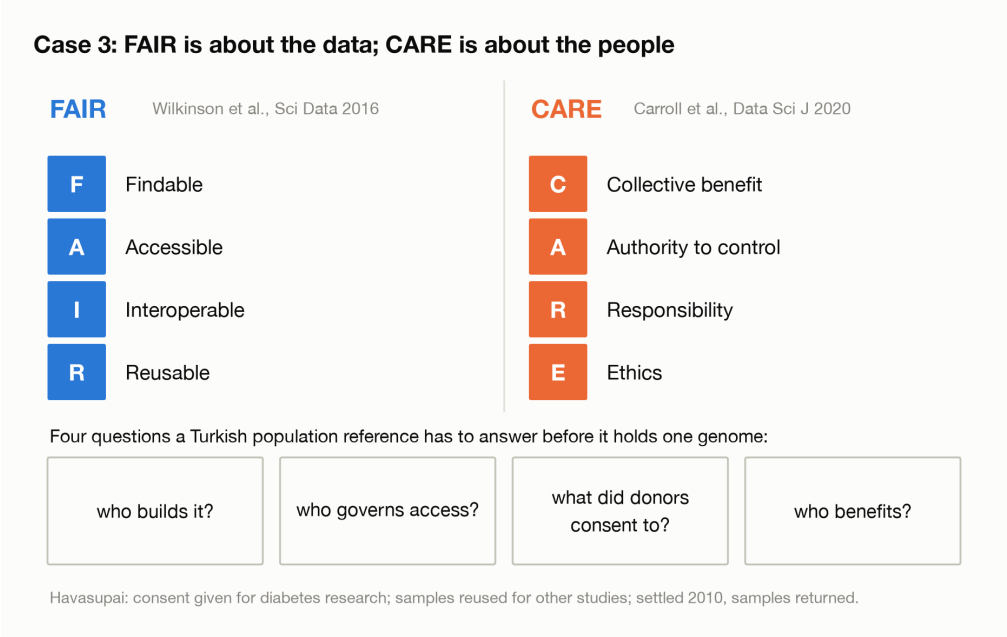
- Week 13: gnomAD v4.1 holds 807,162 people; its Middle Eastern group is in the low thousands
- So the frequency criteria PM2, BS1 and BA1 often cannot fire for a Turkish patient: more VUS
- The systemic fix is a Turkish reference: TÜSEB/TÜGEM and the Türkiye Genom Projesi
- Havasupai: blood given for diabetes research reused for other studies without consent (settled 2010)
- Question: whose job is it to close the gap — and who should own and govern the reference?




*Case 3: the frequency criteria need the patient's own population*

# Case 3 · What a good answer does

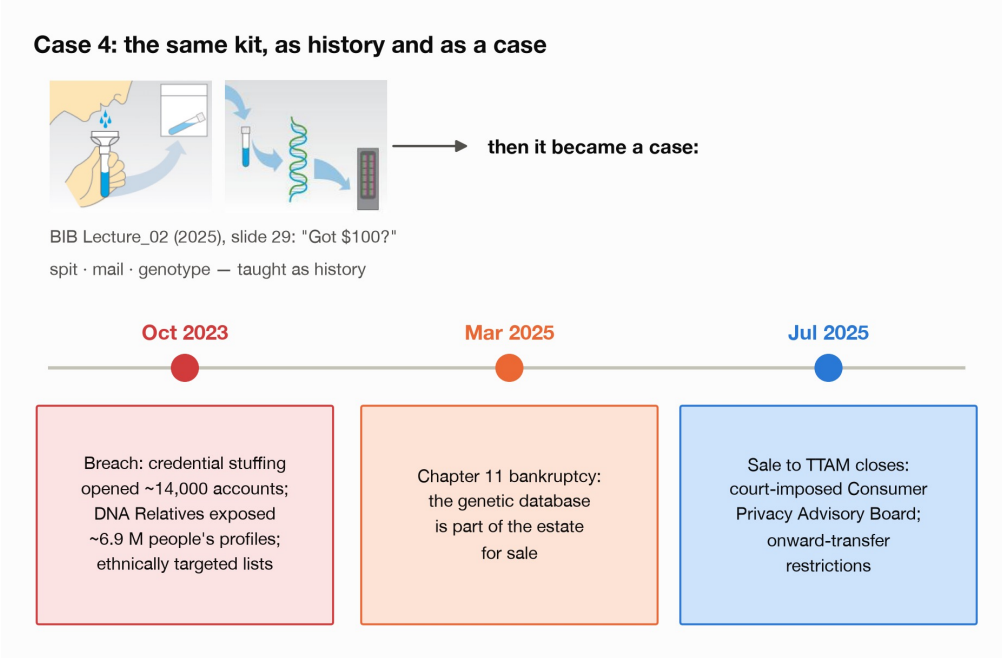
- Separates who builds it (national infrastructure) from who governs access and reuse
- Fixes the consent scope (rıza kapsamı) up front: named uses or a re-consent route
- Puts CARE beside FAIR: FAIR makes data reusable; CARE asks who benefits, who decides
- Tiers access: genomes controlled; even summary frequencies are a decision (Homer 2008)
- Meanwhile, reports which ancestry group each frequency came from, or that it is missing
- No single right owner: a markable line names a governor, a consent scope and an access tier




*FAIR is about the data; CARE is about the people*

# Case 4 · 23andMe: from a $100 kit to a bankruptcy sale

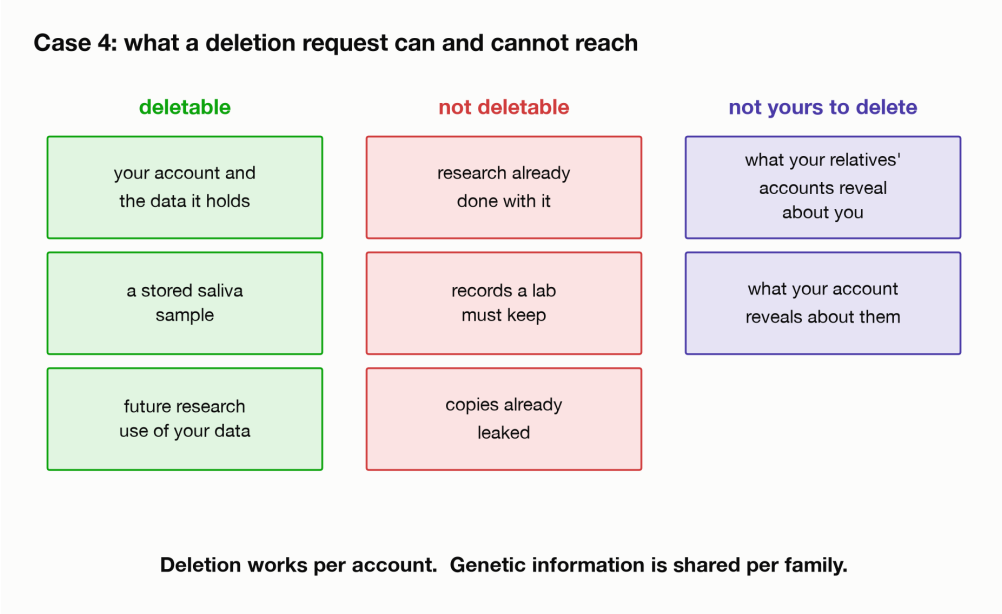
- Last year's deck taught it as history: "Got $100?" — spit in a tube, mail it, read a report online
- Oct 2023: credential-stuffing breach; DNA Relatives exposed millions; ethnically targeted lists posted
- March 2025: Chapter 11 bankruptcy (iflas) — the customers' genetic data is part of the assets for sale
- July 2025: sale to TTAM closes; court-imposed privacy advisory board and onward-transfer limits
- Question: a friend asks, "Should I delete my 23andMe data?" What can be deleted — and what cannot?




*The 2025 deck's "Got $100?" illustrations (BIB Lecture_02, slide 29), then the case*

# Case 4 · What a good answer does

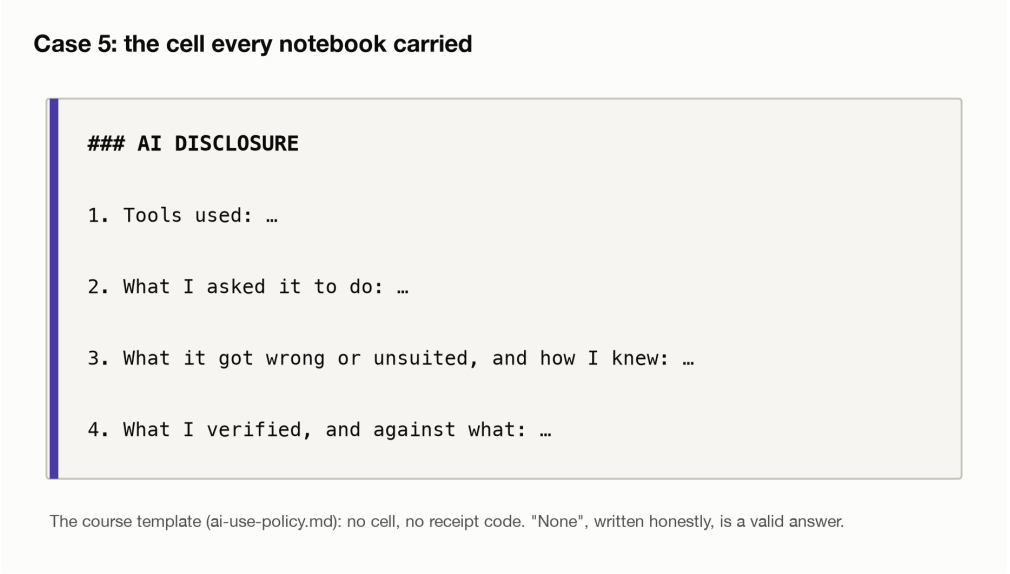
- Deletable: the account, a stored saliva sample, and future research use of your data
- Not deletable: research already done with it, records a lab must keep, copies already leaked
- Not yours to delete: what your relatives' accounts reveal about you — never only yours, again
- Knows where protection came from: court conditions and state attorneys general, not the policy
- A privacy policy is an asset of the bankruptcy estate (iflas masası): it transfers with the data
- Methods point: deletion works per account, but genetic information is shared per family




*What a deletion request can and cannot reach*

# Case 5 · Your AI DISCLOSURE cell, rewritten for a journal

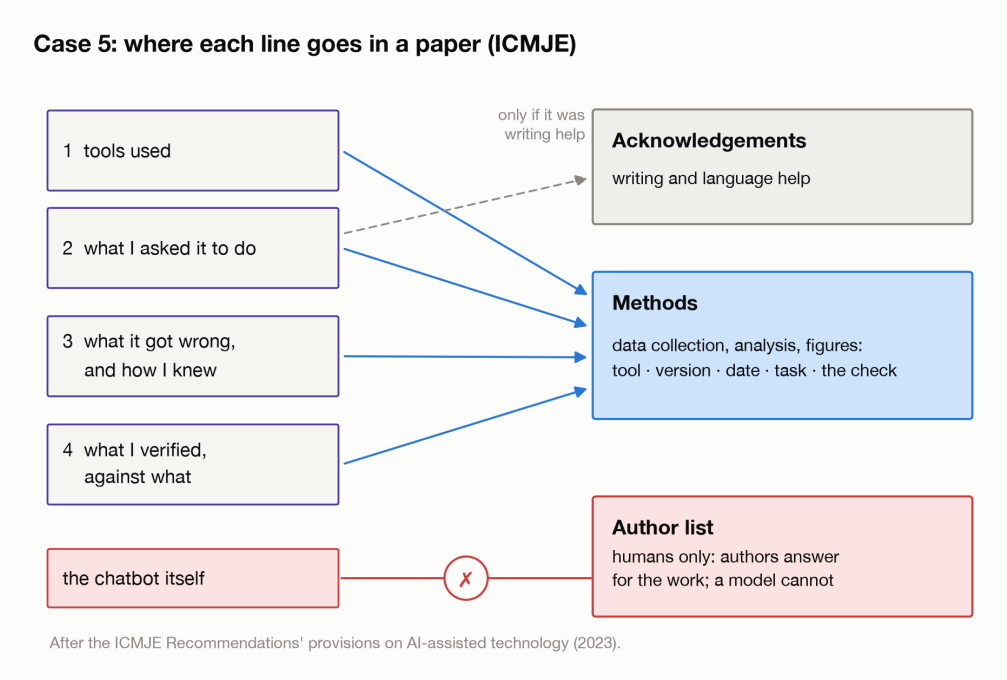
- Every notebook carried a ### AI DISCLOSURE cell (YZ kullanım beyanı): no cell, no receipt code
- Four lines: tools used · what I asked · what it got wrong, and how I knew · what I verified, against what
- BLUE tier: in the Design Notebooks, AI use was required and the record was part of the submission
- Now your Design Notebook 1 becomes a short paper, with your name on it as an author
- Question: write the ICMJE version of your disclosure. Where does each of your four lines go?




*The cell every notebook carried — the course template*

# Case 5 · What a good answer does

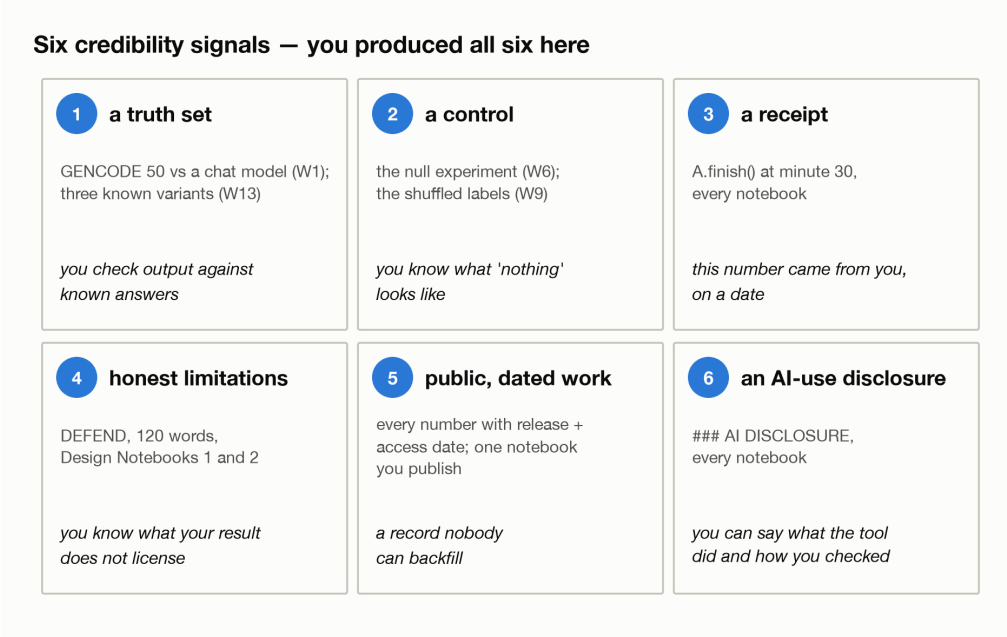
- Keeps the chatbot off the author list: authors must answer for the work, and a model cannot (ICMJE)
- Puts writing and language help in the Acknowledgements (teşekkür)
- Puts use in data collection, analysis or figures in the Methods (yöntem): tool, version, date, task
- Keeps line 3 — what it got wrong, and how you knew: the most useful sentence in your Methods
- Rejects "we used ChatGPT to help": a reviewer cannot act on it
- Signs for every line, including the ones a model wrote: your name is the responsibility (sorumluluk)




*Where each line of the cell goes in a paper — after the ICMJE Recommendations*

# Careers: the mechanism, not the doom

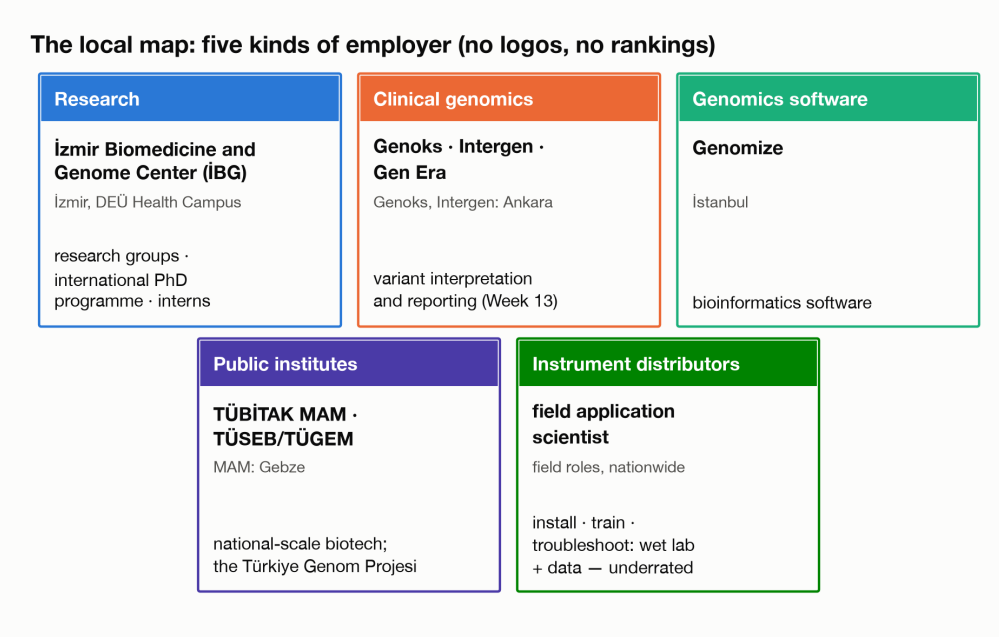
- AI substitutes for codified knowledge (kodlanmış bilgi): what is written down and testable
- It complements tacit knowledge (örtük bilgi): judgement you only get by doing the work
- Entry-level work was disproportionately codified, so that is where the squeeze lands first
- What still signals credibility is evidence of judgement — and this course made you produce six:
  - a truth set · a control · a receipt · honest limitations · public, dated work · an AI-use disclosure




*Six credibility signals, and where each one was made in this course*

# The local map: who hires a bioengineer who can verify

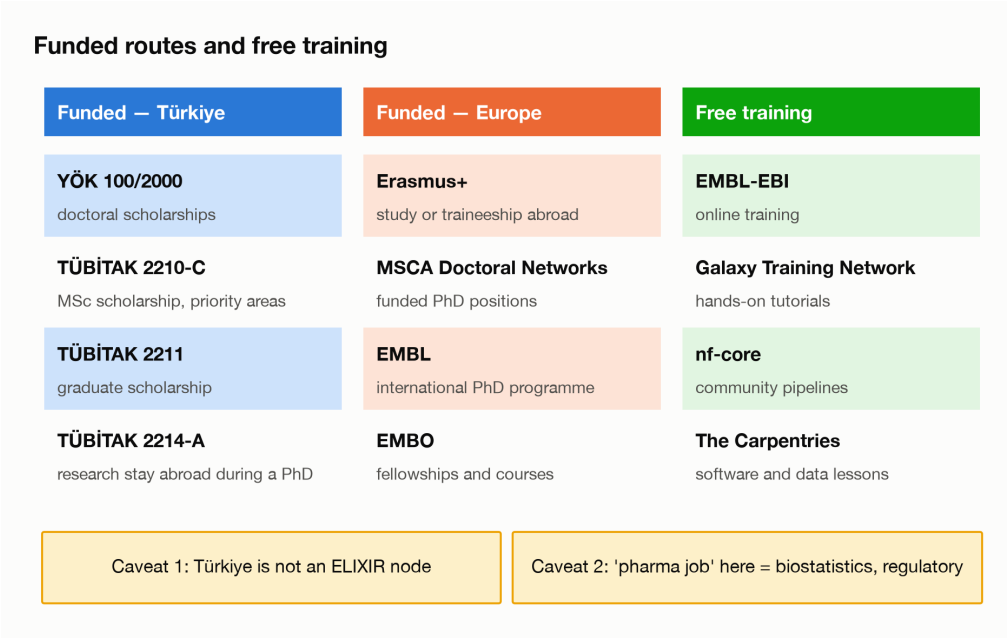
- İzmir Biomedicine and Genome Center (İBG), in this city: research groups, a PhD programme, interns
- Clinical genomics: Genoks, Intergen, Gen Era — and genomics software: Genomize
- Public institutes: TÜBİTAK MAM; TÜSEB/TÜGEM and the Türkiye Genom Projesi
- Field application scientist (saha uygulama uzmanı) at the Turkish instrument distributors
  - the most underrated fit for a bioengineer: wet lab and data, in front of the customer




*The local map: five kinds of employer — no logos, no rankings*

# Funded routes, free training, two caveats

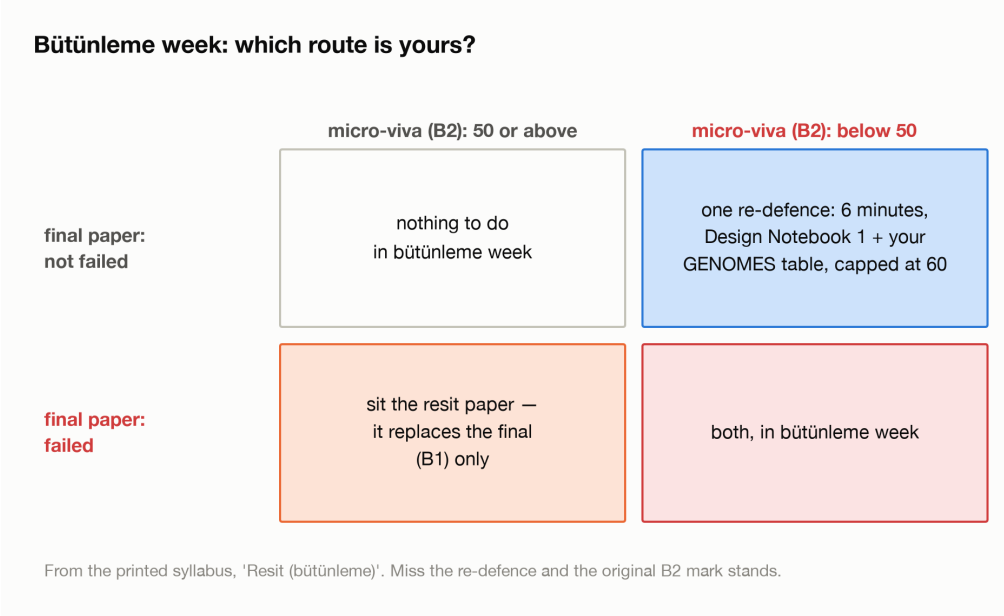
- In Türkiye: YÖK 100/2000 doctoral scholarships; TÜBİTAK 2210-C, 2211, 2214-A
- In Europe: Erasmus+, MSCA Doctoral Networks, EMBL and EMBO
- Free, open now: EMBL-EBI training, the Galaxy Training Network, nf-core, The Carpentries
- Caveat 1: Türkiye is not an ELIXIR node — check eligibility before you plan around ELIXIR
- Caveat 2: a "pharma job" here mostly means biostatistics and regulatory work




*Funded routes and free training*

# Bütünleme: the re-defence route, read from the policy

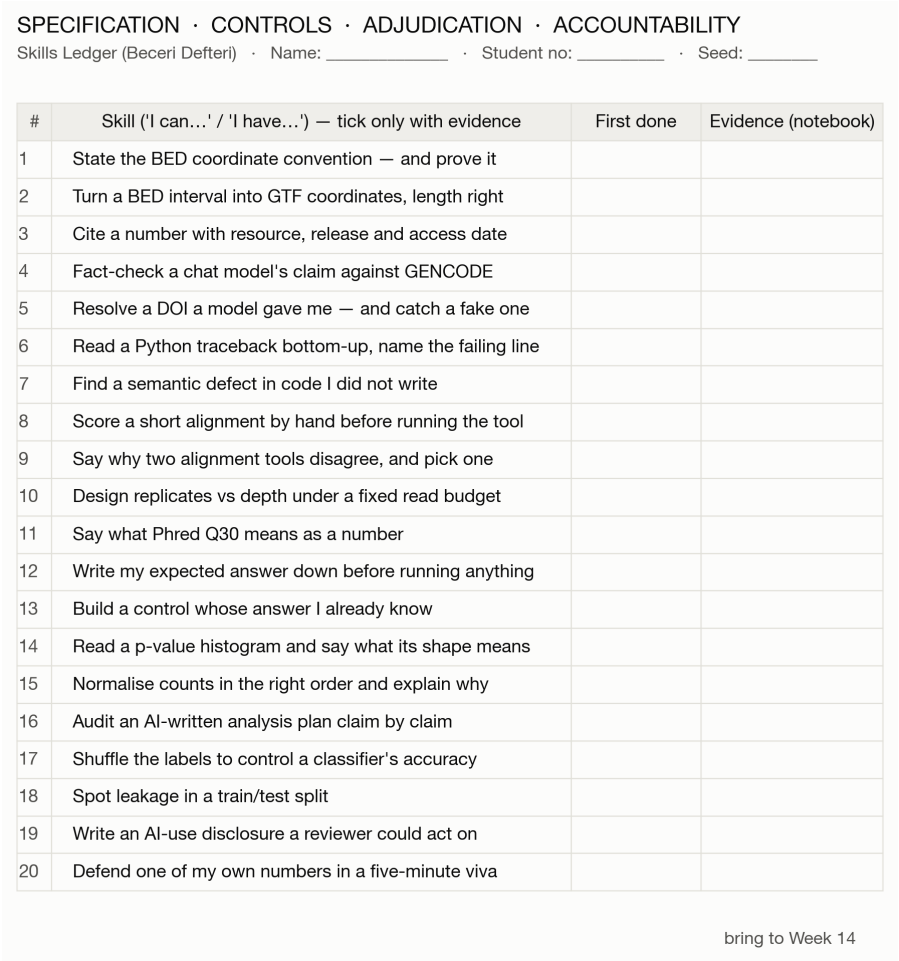
- Bütünleme replaces the final exam (B1, 34%) only; A1, A2 and A3 marks stand
- Below 50 on the micro-viva (B2): exactly one re-defence (yeniden savunma), in bütünleme week
- In a slot published with the resit timetable, on your Design Notebook 1 and latest GENOMES table
- Six minutes, three questions, the same rubric card, a different number drawn at random
- The re-defence mark replaces B2, capped at 60; clearing 50 lifts the hurdle (baraj) cap
- Fail the paper only: resit paper. Fail the hurdle only: re-defence. Fail both: both, that week.




*Bütünleme week: which route is yours?*

# This hour: the last carousel cohort and your Skills Ledger

- No laptop. The final carousel cohort is called from minute 3 — the list was published in Week 12
- Everyone else: the Skills Ledger (Beceri Defteri), one page, three columns
  - requirement (from the fourteen Ads) · my evidence (a notebook, a number, a defect) · gap
- A TA or a peer pair signs it off in two minutes: three yes/no checks — recorded complete, not scored
- Then one A4 side: one sentence on what you can now verify that you could not in September, plus one gap
- The A4 is collected at minute 30 and handed back to you at the final



SPECIFICATION · CONTROLS · ADJUDICATION · ACCOUNTABILITY

Skills Ledger (Beceri Defteri) · Name: _____ · Student no: _____ · Seed: _____

| # | Skill ('I can…' / 'I have…') — tick only with evidence | First done | Evidence (notebook) |
|---|---|---|---|
| 1 | State the BED coordinate convention — and prove it | | |
| 2 | Turn a BED interval into GTF coordinates, length right | | |
| 3 | Cite a number with resource, release and access date | | |
| 4 | Fact-check a chat model's claim against GENCODE | | |
| 5 | Resolve a DOI a model gave me — and catch a fake one | | |
| 6 | Read a Python traceback bottom-up, name the failing line | | |
| 7 | Find a semantic defect in code I did not write | | |
| 8 | Score a short alignment by hand before running the tool | | |
| 9 | Say why two alignment tools disagree, and pick one | | |
| 10 | Design replicates vs depth under a fixed read budget | | |
| 11 | Say what Phred Q30 means as a number | | |
| 12 | Write my expected answer down before running anything | | |
| 13 | Build a control whose answer I already know | | |
| 14 | Read a p-value histogram and say what its shape means | | |
| 15 | Normalise counts in the right order and explain why | | |
| 16 | Audit an AI-written analysis plan claim by claim | | |
| 17 | Shuffle the labels to control a classifier's accuracy | | |
| 18 | Spot leakage in a train/test split | | |
| 19 | Write an AI-use disclosure a reviewer could act on | | |
| 20 | Defend one of my own numbers in a five-minute viva | | |

bring to Week 14

*The ledger you were given in Week 1: today it becomes requirement / my evidence / gap*